EXP.NO: 5 DATE:05/03/2025	Write a Servlet to demonstrate the difference between HTTP GET and POST methods by creating a form and handling requests accordingly.

AIM:

To develop an Online Quiz Application using HTML and Java Servlet to demonstrate handling of HTTP GET and POST methods, where GET is used to fetch quiz questions and POST is used to submit answers.

Algorithm :

1. Input:
- The user accesses the quiz page.
- The user answers multiple-choice questions.
- The form submits responses using GET and POST methods.

2. Steps for Creating and Handling the Application:

Step 1: Create an HTML Quiz Form
- Design an HTML page with:
  - Multiple-choice questions.
  - Radio buttons for answers.
- Use:
  - GET method to load or view quiz questions.
  - POST method to submit answers.

Step 2: Create the Servlet
- Create a Java Servlet class.
- Implement:
  - doGet() → to display quiz questions.
  - doPost() → to evaluate answers.

Step 3: Handle GET Request
- In doGet():
  - Display quiz questions dynamically.
  - Send HTML form as response.

Step 4: Handle POST Request
- In doPost():
  - Retrieve answers using request.getParameter().
  - Compare answers with correct ones.
  - Calculate score.

Step 5: Display Result
- Show:
  - Score
  - Correct/incorrect answers

Step 6: Deploy and Test
- Compile servlet.
- Deploy in Apache Tomcat.
- Test:
  - Loading quiz (GET)
  - Submitting quiz (POST)

Step 7: Observe Difference
- GET → displays quiz (data visible in URL).
- POST → submits answers (data hidden in request body).

CODE :

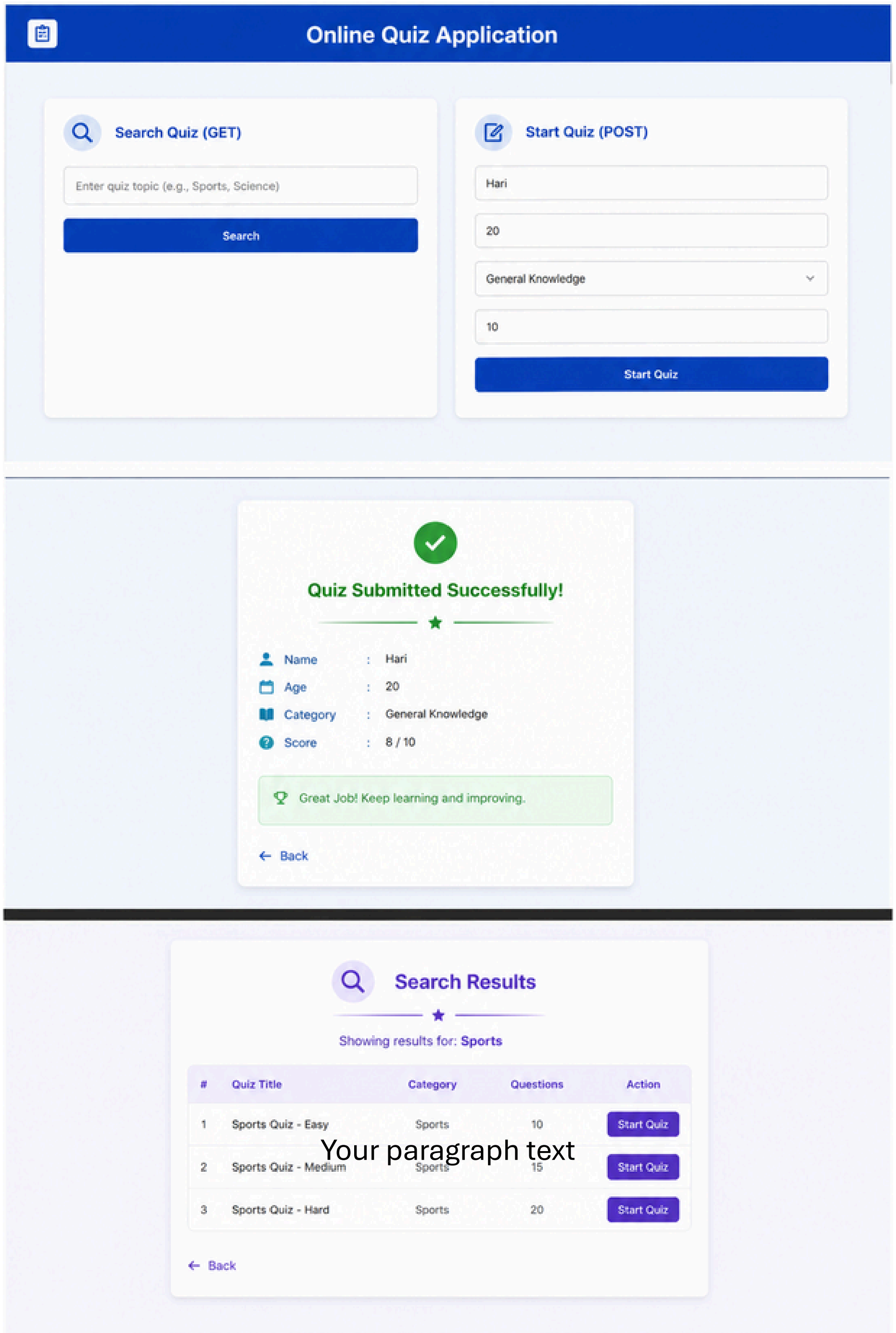
Quiz.html

```html
<!DOCTYPE html>
<html>
<head>
   <title>Online Quiz</title>
</head>
<body style="font-family:Segoe UI;background:#f2f6ff;">

<h2 align="center">Online Quiz Application</h2>

<form action="QuizServlet" method="post">

<p>1. What is Java?</p>
<input type="radio" name="q1" value="a"> Programming Language<br>
<input type="radio" name="q1" value="b"> Operating System<br>

<p>2. Which is used for web development?</p>
<input type="radio" name="q2" value="a"> HTML<br>
<input type="radio" name="q2" value="b"> MS Word<br>

<p>3. Which method is secure?</p>
<input type="radio" name="q3" value="a"> GET<br>
<input type="radio" name="q3" value="b"> POST<br>

<br>
<button type="submit">Submit Quiz</button>

</form>

</body>
</html>
```

QuizServlet.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;

@WebServlet("/QuizServlet")
public class QuizServlet extends HttpServlet {

  protected void doGet(HttpServletRequest request, HttpServletResponse response)
      throws ServletException, IOException {

    response.setContentType("text/html");
    PrintWriter out = response.getWriter();

    out.println("<html><body>");
    out.println("<h2>Quiz Loaded Successfully (GET Method)</h2>");
    out.println("<a href='Quiz.html'>Start Quiz</a>");
    out.println("</body></html>");
  }

  protected void doPost(HttpServletRequest request, HttpServletResponse response)
      throws ServletException, IOException {

    response.setContentType("text/html");
    PrintWriter out = response.getWriter();

    String q1 = request.getParameter("q1");
    String q2 = request.getParameter("q2");
    String q3 = request.getParameter("q3");

    int score = 0;

    if ("a".equals(q1)) score++;
    if ("a".equals(q2)) score++;
    if ("b".equals(q3)) score++;

    out.println("<html><body style='font-family:Segoe UI;background:#f2f6ff;'>");
    out.println("<div style='width:400px;margin:50px
auto;background:white;padding:20px;border-radius:10px;'>");
    out.println("<h2>Quiz Result</h2>");
    out.println("<p>Your Score: <b>" + score + "/3</b></p>");
    out.println("<br><a href='Quiz.html'>Try Again</a>");
    out.println("</div></body></html>");
  }
}
```

OUTPUT :



Result :

The Notice Board interface is successfully converted into an Online Quiz Application, maintaining the same UI structure while demonstrating GET (load quiz) and POST (submit answers) functionality.